

Evolution of Computers & Modern Computer Systems

1. History & Evolution of Computers

From ancient times to the present, humans have experimented with various devices for mathematical calculations and other tasks. Passing through several eras, computer technology has reached a highly advanced state today.

Major Eras of Computer Evolution

1. **Pre-Mechanical Era** (Before 1450)
2. **Mechanical Era** (1450 – 1840)
3. **Electro-Mechanical Era** (1840 – 1940)
4. **Electronic Era** (1940 to Present)

1. Pre-Mechanical Era (Before 1450)

- **Abacus:** Created by the Chinese around 3000 BC for calculations. It is a counting frame using moveable beads.

2. Mechanical Era (1450 – 1840)

- **Napier's Bones (1617):** Invented by John Napier. It used ivory rods marked with numbers to perform logarithmic and multiplication calculations.
- **Pascaline / Adding Machine (1642):** Created by Blaise Pascal to perform addition and subtraction mechanically using gears and wheels.
- **Stepped Reckoner (1674):** Introduced by Gottfried Wilhelm Leibniz. It could perform addition, subtraction, multiplication, and division.
- **Charles Babbage's Difference Engine (1822):** Considered the first mechanical computer. Introduced the fundamental input/output concept. For his groundbreaking work, Charles Babbage is known as the "**Father of the Computer**".
- **Analytical Engine (1833):** Designed by Charles Babbage using the punched card concept introduced by Joseph Marie Jacquard (used in automated looms). Included components for Data Input, Storage, Processing, and Output.
- **First Computer Programmer:** Ada Augusta Lovelace developed the algorithms/programs for the Analytical Engine, making her the world's first computer programmer.

3. Electro-Mechanical Era (1840 – 1940)

- **Electronic Valve / Vacuum Tube (1906):** Invented by Lee De Forest. Later used as the core switching element in 1st Generation computers.
- **Harvard Mark I (1939):** Built by Howard Aiken in collaboration with IBM. Recognized as the first automatic digital electro-mechanical computer. Weighed around 5 tons.

2. The Electronic Era & Computer Generations

Computers developed from 1940 onwards are classified into 5 distinct generations based on technology:

Generation	Primary Technology	Key Characteristics	Example Computers
1st Gen (1940–1956)	Vacuum Tubes	Very large size, high heat generation, high power consumption, low reliability, expensive.	ABC, ENIAC, EDSAC, EDVAC, UNIVAC-1
2nd Gen (1956–1963)	Transistors	Smaller, produced less heat, consumed less power, more reliable and cheaper than 1st Gen.	IBM 1620, IBM 7094, CDC 1604, CDC 3600, UNIVAC 1108
3rd Gen (1963–1975)	Integrated Circuits (ICs)	Higher reliability, compact size, low power, portable, used high-level programming languages.	IBM 360 Series, PDP (Personal Data Processor), TDC 316
4th Gen (1975–1989)	Microprocessors (VLSI - Very Large Scale Integration)	Extremely reliable, very small size, low power consumption, personal computing era.	IBM PC, STAR 1000, PDP 11
5th Gen (1989–Present)	Microprocessors (ULSI - Ultra Large Scale Integration)	Ultra-high reliability, minimal power use, integration of Artificial Intelligence (AI) & parallel processing.	Desktops, Laptops, Notebooks, Ultrabooks

Details of Key 1st Generation Computers:

- **ABC (Atanasoff-Berry Computer - 1942):** Built by John Vincent Atanasoff and Clifford E. Berry. Recognized as the first electronic digital calculating device.
- **ENIAC (Electronic Numerical Integrator and Computer - 1946):** Developed by John Mauchly and J. Presper Eckert. Used ~18,000 vacuum tubes, capable of 5,000 operations/sec. Generated intense heat requiring frequent valve replacements.
- **EDSAC (Electronic Delay Storage Automatic Calculator - 1947):** Built by Maurice Wilkes based on John von Neumann's architecture. First stored-program computer in practical operation.
- **EDVAC (Electronic Discrete Variable Automatic Computer - 1948):** Designed by John Mauchly and J. Presper Eckert. Utilized binary stored-program concepts.
- **UNIVAC-1 (Universal Automatic Computer - 1950):** Built by Remington Rand. First commercially produced digital computer; used thousands of vacuum tubes and punch cards for I/O.

3. Classification of Computers

Computers can be categorized based on three main criteria:

2. By Purpose

- **Special Purpose Computers:** Designed to perform specific task (e.g., medical imaging machines, traffic control).
- **General Purpose Computers:** Designed to perform a wide variety of daily applications (e.g., PCs, laptops).

1. By Technology

- **Analog Computers:** Process continuous physical data (pressure, temperature, voltage).
- **Digital Computers:** Process discrete binary data (0s and 1s). High efficiency and precision.
- **Hybrid Computers:** Combine analog and digital features (e.g., accepting analog inputs and displaying digital output like in medical ICU monitors).

3. By Size & Performance

- **Supercomputers:** Extremely fast, massive memory, very expensive, large physical footprint. Used in military applications, weather forecasting, and scientific research.
- **Mainframe Computers:** Large-scale systems serving as central hubs for thousands of connected users simultaneously. Less expensive than supercomputers. Used in large enterprises.
- **Minicomputers:** Medium-sized multi-user servers, commonly used as network servers in banks and financial institutions.
- **Microcomputers:** Powered by microprocessors, compact and cost-effective for daily use:
 - *Desktop Computers:* Fits on top of a desk.
 - *Laptop / Notebook:* Portable computers with integrated screen/keyboard.
 - *Palmtop / Smart Phone:* Handheld computing devices with specialized OS.
 - *Tablet / Phablet:* Touchscreen portable devices filling the gap between phones and laptops.

4. Hardware Components & Computer Functionality

Hardware refers to all physical components of a computer system. They are classified as follows:

A. Input Devices

Used to feed data and instructions into the computer system.

- **Standard Keyboards:** Primary text and command entry device.
- **Pointing Devices:** Mouse, Touch Pad, Joystick, Light Pen, Graphic Tablet, Touch Screen.
- **Direct Entry Input Devices:**
 - Barcode Reader, Magnetic Stripe Reader (credit cards), Smart Card Reader.
 - Digital Camera, Video Camera, Web Camera, CCTV.
 - Microphone (Audio input).
 - Optical Mark Reader (OMR), Optical Character Reader (OCR), Magnetic Ink Character Recognition (MICR).
 - Sensors: Water level, electricity, light (LDR), sound, current sensors.

Advantages of Direct Entry Input Devices vs. Keyboards:

1. Automated data capture minimizes operational costs.

2. Significantly reduces data entry errors.
3. Saves time by capturing data instantly.

B. Output Devices

Converts processed data into human-readable information.

- **Display Screens (Monitors):** Images rendered via small dots called pixels.
 - *Cathode Ray Tube (CRT):* Electron beams fire onto a phosphor-coated screen.
 - *Liquid Crystal Display (LCD):* Uses Thin Film Transistors (TFT) for each pixel.
 - *Light Emitting Diode (LED):* Modern flat panel technology using liquid crystals with LED backlighting.

LED Advantages: Low cost, highly reliable, minimal power usage, long lifespan, eco-friendly, high dynamic contrast ratio.
- **Printers:** Converts soft copies into hard physical copies.
 - *Impact Printers:* Physical impact prints characters, noisy (e.g., **Dot Matrix Printer** used in offices for carbon/continuous forms).
 - *Non-Impact Printers:* Quiet, high quality.
 - **Inkjet Printer:** Sprays tiny ink droplets onto paper via print head nozzles. Uses ink cartridges.
 - **Laser Printer:** Uses laser beam and toner powder for high-speed, high-quality printing.

C. Storage Devices

Used to retain data and software permanently or temporarily.

- **Magnetic Storage:**
 - *Fixed Internal Hard Disk (HDD):* Primary magnetic storage permanently installed.
 - *Portable External Hard Disk:* Connects via USB for backups.
 - *Magnetic Tape:* High-capacity (up to Terabytes) sequential access backup medium.
- **Optical Disks:** Read/written using low-power laser beams (Random Access):

Type	Characteristics	Capacity
CD-ROM / CD-R / CD-RW	Read-only / Write once / Rewritable	650 – 900 MB
DVD-ROM / DVD-R / DVD-RW	Read-only / Write once / Rewritable	4.7 – 9.4 GB
Blu-ray (BD / RE / 3D)	High-definition & 3D video storage / Rewritable	25 – 128 GB

- **Flash Memory:** Solid-state transistor storage (e.g., USB Pen Drives, Memory Cards, SSDs).

5. System Architecture, Processing & Memory

The Motherboard & System Bus

The motherboard is the main printed circuit board connecting internal components (CPU, RAM, ROM, Hard Disk) and external peripherals. Communication is handled by signal buses:

- **Data Bus:** Transfers actual data between CPU, memory, and peripherals.
- **Address Bus:** Carries memory address signals indicating where data should be read from or written to.
- **Control Bus:** Transmits control signals and timing commands from the CPU to regulate components.

Chipset Architecture:

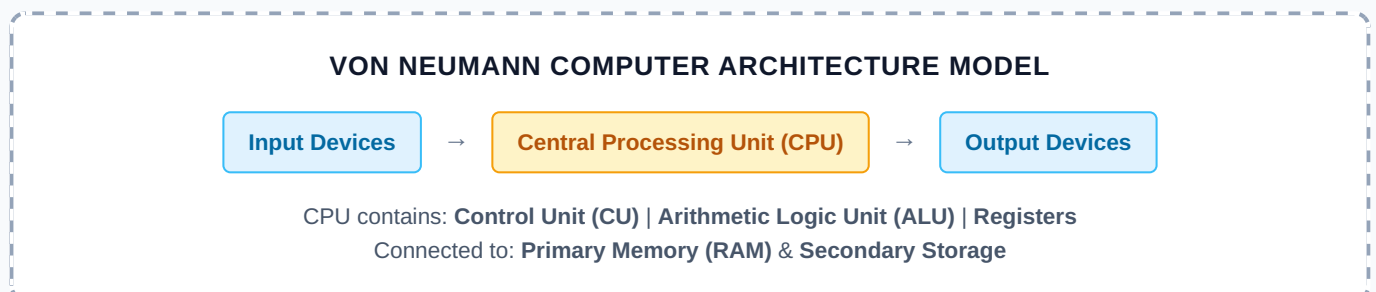
- **Northbridge:** High-speed controller connecting CPU, RAM, and graphics bus.
- **Southbridge:** Manages slower I/O buses (ISA, PCI, USB, audio, storage interfaces).

Advanced Computing Paradigms

- **Parallel Computing:** Splits a complex computational problem into smaller parts that are executed simultaneously across multiple processing units, producing a unified output.
- **Grid Computing:** Combines geographically distributed computer resources across networks to solve large-scale computational problems independently.

6. John von Neumann Architecture

Proposed by John von Neumann, this model forms the foundation of all modern general-purpose digital computers:



Key Principles of Von Neumann Architecture:

1. Data and instructions are stored together in the same primary memory unit.
2. Memory locations are addressed by their numeric locations, independent of the content stored.
3. Execution of instructions occurs sequentially unless explicitly altered.
4. Consists of distinct Processing Unit (ALU/Registers), Control Unit, Memory, and I/O mechanisms.

The Fetch-Execute Cycle (Machine Cycle)

Managed continuously by the Operating System when programs run:

1. Fetch Instruction



2. Decode Instruction



3. Execute Instruction



4. Store Result

Multicore Processors

Single-core CPUs execute only one instruction at a time. Modern CPUs integrate multiple physical/logical processing cores into a single physical chip (e.g., Dual-Core [i3], Quad-Core [i5], Hexa/Octa-Core [i7]) to enable true parallel instruction execution, faster execution, and maximum efficiency.

7. Memory Classification & Hierarchy

Memory Hierarchy Comparison Table

Memory Type	Physical Size	Capacity	Cost / Bit	Access Speed
CPU Registers	Smallest	Very Low	Highest	Fastest
Cache Memory	Very Small	Low	Very High	Ultra Fast
Primary Memory (RAM)	Medium	Medium	Moderate	Fast
Flash Memory / SSD	Compact	High	Low	Moderate/Slow
Hard Disk Drive	Large	Very High	Very Low	Slow
Magnetic Tape	Large	Massive	Lowest	Very Slow (Sequential)

Volatile vs. Non-Volatile Memory

Volatile Memory (Temporary)

Loses all stored data immediately when power is switched off.

- **CPU Registers:** Located inside the CPU; ultra-fast temporary storage.
- **Cache Memory:** High-speed buffer between CPU and RAM. Stores frequently used instructions.
 - **L1 Cache:** Built into CPU, fastest, smallest capacity.
 - **L2 Cache:** Larger than L1, slightly slower.
 - **L3 Cache:** Largest cache, shared across cores, doubles effective RAM response speed.

Non-Volatile Memory (Permanent)

Retains stored data permanently even when electrical power is disconnected.

- **ROM (Read Only Memory):** Stores firmware / BIOS boot code.
 - **PROM:** Programmable once by user.
 - **EPROM:** Erasable using Ultraviolet (UV) light exposure.
 - **EEPROM:** Electrically erasable and reprogrammable.
- **Secondary Storage:** Holds user files long-term (HDDs, SSDs, Optical Media, Tapes).

- **RAM (Random Access Memory):** Primary workspace for active OS and programs.
 - *SRAM (Static RAM):* Uses transistors, no refresh required, faster, used for Cache.
 - *DRAM (Dynamic RAM):* Uses capacitors, requires continuous electrical refresh, high density.
 - *SDRAM (Synchronous DRAM):* Synchronized with system clock speed (DDR, DDR2, DDR3, DDR4).

Memory Access Methods

- **Sequential Access:** Data must be read in linear order, one record after another (e.g., Magnetic Tape).
- **Random Access (Direct Access):** Data can be accessed directly from any memory address in constant time (e.g., RAM, Hard Disk, Flash Storage).